Subject: Mathematics | **PERIODIC TEST – 04 (PROBABILITY)** | **Time:** 60 Minutes
Class: Class X (CBSE) | **Chapter 14: Probability** | **Max. Marks:** 30

SECTION A: OBJECTIVE & MCQs (1 Mark Each) [5 Marks]

Question 1 [1 Mark]

Three identical unbiased dice are rolled together. The probability that the same number appears on all three dice is:

(a) $\frac{1}{36}$ (b) $\frac{1}{216}$
(c) $\frac{1}{6}$ (d) $\frac{5}{36}$

Working & Calculation:

- When three dice are rolled simultaneously, total elementary outcomes:

$$n(S) = 6 \times 6 \times 6 = 216$$

- The outcomes where the same number appears on all three dice (triplets) are:

$$E = \{(1,1,1), (2,2,2), (3,3,3), (4,4,4), (5,5,5), (6,6,6)\} \implies n(E) = 6$$

- Probability:

$$P(\text{same number on all three dice}) = \frac{n(E)}{n(S)} = \frac{6}{216} = \frac{1}{36}$$

CORRECT ANSWER: OPTION (a) $\longrightarrow \frac{1}{36}$

Question 2 [1 Mark]

From the numbers 3, 5, 5, 7, 7, 7, 9, 9, 9, 9, one number is selected at random. The probability that the selected number is their mean is:

(a) $\frac{3}{10}$ (b) $\frac{1}{10}$
(c) $\frac{7}{10}$ (d) $\frac{2}{5}$

Working & Calculation:

- Total given numbers: $n(S) = 1 + 2 + 3 + 4 = 10$ observations.
- Sum of observations:

$$\Sigma x = 3(1) + 5(2) + 7(3) + 9(4) = 3 + 10 + 21 + 36 = 70$$

- Arithmetic Mean:

$$\bar{x} = \frac{\Sigma x}{n} = \frac{70}{10} = 7$$

- Frequency of the mean value 7 in the dataset is 3 (since 7 appears three times) $\implies n(E) = 3$.

$$P(\text{selected number is mean}) = \frac{n(E)}{n(S)} = \frac{3}{10}$$

CORRECT ANSWER: OPTION (a) $\longrightarrow \frac{3}{10}$

Question 3 [1 Mark]

A lot of 20 bulbs contains 4 defective ones. One bulb is drawn at random from the lot. The probability that this bulb is non-defective is:

(a) $\frac{4}{5}$ (b) $\frac{1}{5}$

(c) $\frac{3}{4}$ (d) $\frac{1}{4}$

Working & Calculation:

- Total bulbs in the lot $n(S) = 20$. Defective bulbs $= 4$.
- Non-defective bulbs $= 20 - 4 = 16$ bulbs.
- Probability:

$$P(\text{non-defective}) = \frac{n(E)}{n(S)} = \frac{16}{20} = \frac{4}{5} = 0.8$$

CORRECT ANSWER: OPTION (a) $\longrightarrow \frac{4}{5} = 0.8$

Question 4 [1 Mark]

The probability of guessing the correct answer to a certain test question is $\frac{x}{12}$. If the probability of not guessing the correct answer is $\frac{2}{3}$, then x is equal to:

(a) **2** (b) **3**

(c) **4** (d) **6**

Working & Calculation:

- Guessing the correct answer (E) and not guessing the correct answer ($\bar{E}$) are complementary events:

$$P(E) + P(\bar{E}) = 1$$

- Substitute given probabilities:

$$\frac{x}{12} + \frac{2}{3} = 1 \implies \frac{x}{12} = 1 - \frac{2}{3} = \frac{1}{3}$$

- Solve for x:

$$x = 12 \times \frac{1}{3} = 4$$

CORRECT ANSWER: OPTION (c) $\longrightarrow$ $x = 4$

Question 5 (Assertion–Reason) [1 Mark]

Assertion (A): The probability that a non-leap year has 53 Tuesdays is $\frac{2}{7}$.
Reason (R): A non-leap year contains 365 days, giving 52 weeks and 1 odd day.

(a) **Both (A) and (R) are true, and (R) is the correct explanation of (A).**
(b) **Both (A) and (R) are true, but (R) is not the correct explanation of (A).**
(c) **(A) is true, but (R) is false.**
(d) **(A) is false, but (R) is true.**

Step-by-Step Logical Analysis:

1. **Reason (R):** 365 days $= (52 \times 7) + 1$ day $= 52$ weeks $+ 1$ extra day. $\implies$ **(R) is TRUE.**
2. **Assertion (A):** 52 complete weeks contain exactly 52 Tuesdays. The 53rd Tuesday occurs only if the single extra day is a Tuesday (1 favourable day out of 7 equally likely weekdays):

$$P(\text{53 Tuesdays in non-leap year}) = \frac{1}{7} \neq \frac{2}{7}$$

   Thus, Assertion (A) claims an incorrect probability of $\frac{2}{7}$. $\implies$ **(A) is FALSE.**
3. **Conclusion:** Assertion (A) is false, but Reason (R) is true.

CORRECT ANSWER: OPTION (d) $\longrightarrow$ (A) is false, but (R) is true.

SECTION B: VERY SHORT ANSWER QUESTIONS (2 Marks Each) [4 Marks]

Question 6 [2 Marks]

A bag contains 4 red, 3 blue, and 2 yellow balls. One ball is drawn at random. Find the probability that the ball drawn is: (i) red or blue (ii) neither yellow nor red.

Sample Space: Total balls $n(S) = 4 \text{ red} + 3 \text{ blue} + 2 \text{ yellow} = 9$ balls.

(i) Red or blue: Favourable balls $= 4 \text{ red} + 3 \text{ blue} = 7 \implies P = \frac{7}{9}$.

Answer (i): $\mathbf{P(\text{red or blue}) = \frac{7}{9}}$

(ii) Neither yellow nor red: Ball must be blue $\implies n(E_2) = 3$ blue $\implies P = \frac{3}{9} = \frac{1}{3}$.

Answer (ii): $\mathbf{P(\text{neither yellow nor red}) = \frac{1}{3}}$

Question 7 [2 Marks]

The probability that it will rain tomorrow is 0.85. What is the probability that it will not rain tomorrow?

Step-by-Step Derivation:

1. Let event E be "it will rain tomorrow" $\implies P(E) = 0.85$.
2. Its complement $\bar{E}$ represents "it will not rain tomorrow".
3. By the property of complementary events:

$$P(E) + P(\bar{E}) = 1 \implies P(\bar{E}) = 1 - P(E) = 1 - 0.85 = 0.15 = \frac{15}{100} = \frac{3}{20}$$

Answer: $\mathbf{P(\text{will not rain tomorrow}) = 0.15 = \frac{3}{20}}$

SECTION C: SHORT ANSWER QUESTIONS (3 Marks Each) [6 Marks]

Question 8 [3 Marks]

A piggy bank contains a hundred 50p coins, fifty Rs. 1 coins, twenty Rs. 2 coins, and ten Rs. 5 coins. If it is equally likely that one of the coins will fall out when the bank is turned upside down, what is the probability that the coin: (i) will be a 50p coin? (ii) will not be a Rs. 5 coin? (iii) will be either a Rs. 1 or a Rs. 2 coin?

Coin Counts & Sample Space: Total coins $n(S) = 100$ (50p) $+ 50$ (Rs. 1) $+ 20$ (Rs. 2) $+ 10$ (Rs. 5) $= 180$.

(i) Will be a 50p coin: Number of 50p coins $= 100 \implies P = \frac{100}{180} = \frac{5}{9}$.

Answer (i): $\mathbf{P(\text{50p coin}) = \frac{5}{9}}$

(ii) Will not be a Rs. 5 coin: Coins other than Rs. $5 = 180 - 10 = 170 \implies P = \frac{170}{180} = \frac{17}{18}$.

Answer (ii): $\mathbf{P(\text{not a Rs. 5 coin}) = \frac{17}{18}}$

(iii) Either a Rs. 1 or a Rs. 2 coin: Favourable coins $= 50 + 20 = 70 \implies P = \frac{70}{180} = \frac{7}{18}$.

Answer (iii): $\mathbf{P(\text{Rs. 1 or Rs. 2 coin}) = \frac{7}{18}}$

Question 9 [3 Marks]

Two dice are thrown at the same time. Find the probability that: (i) the difference of the numbers on the two dice is 2 (ii) the difference of the numbers on the two dice is 0 (iii) the number on the first die is greater than 4.

Sample Space: Two dice thrown simultaneously $\implies n(S) = 6 \times 6 = 36$ elementary outcomes.

(i) Difference of numbers is 2: $|x - y| = 2 \implies \{(1,3),(3,1),(2,4),(4,2),(3,5),(5,3),(4,6),(6,4)\}$ (8 pairs):

$$P(\text{difference is } 2) = \frac{8}{36} = \frac{2}{9}$$

Answer (i): $\mathbf{P(\text{difference is } 2) = \frac{2}{9}}$

(ii) Difference of numbers is 0: $|x - y| = 0 \iff x = y$: $\{(1,1),(2,2),(3,3),(4,4),(5,5),(6,6)\}$ (6 pairs):

$$P(\text{difference is } 0) = \frac{6}{36} = \frac{1}{6}$$

Answer (ii): $\mathbf{P(\text{difference is } 0) = \frac{1}{6}}$

(iii) Number on the first die is greater than 4: First die is 5 or 6 ($2 \times 6 = 12$ outcomes):

$$P(\text{first die } > 4) = \frac{12}{36} = \frac{1}{3}$$

Answer (iii): $\mathbf{P(\text{first die } > 4) = \frac{1}{3}}$

SECTION D: LONG ANSWER / HOTS (5 Marks Each) [5 Marks]

Question 10 [5 Marks]

A box contains 12 balls out of which x are black. (i) If one ball is drawn at random from the box, what is the probability that it will be a black ball? (ii) If 6 more black balls are put in the box, the probability of drawing a black ball is now double of what it was before. Find the value of x. (iii) With this value of x, find the probability of drawing a non-black ball from the final box. (iv) If 2 white balls are now removed from the box, find the new probability of drawing a black ball.

Initial State: Total balls $= 12$, black balls $= x$, non-black balls $= 12 - x$.

(i) Probability of drawing a black ball initially: $P_1(\text{black}) = \frac{x}{12}$.

Answer (i): $\mathbf{P_1(\text{black}) = \frac{x}{12}}$

(ii) Value of x after adding 6 black balls: New total $= 18$, new black $= x + 6 \implies P_2 = \frac{x+6}{18}$.

Given $P_2 = 2 \times P_1 \implies \frac{x+6}{18} = 2\left(\frac{x}{12}\right) = \frac{x}{6} \implies x + 6 = 3x \implies 2x = 6 \implies x = 3$.

Answer (ii): $\mathbf{x = 3}$

(iii) Probability of drawing a non-black ball from the final box: Total balls $= 18$, black $= 3 + 6 = 9$, non-black $= 18 - 9 = 9 \implies P = \frac{9}{18} = \frac{1}{2} = 0.5$.

Answer (iii): $\mathbf{P(\text{non-black}) = \frac{1}{2} = 0.5}$

(iv) Probability after removing 2 white balls: New total balls $= 18 - 2 = 16$. Black balls remain $9 \implies P = \frac{9}{16}$.

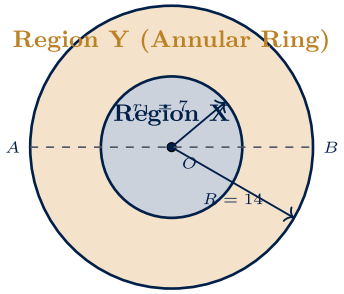
Answer (iv): $\mathbf{P(\text{black ball}) = \frac{9}{16}}$

SECTION E: APPLIED CASE STUDY (4 Marks) [4 Marks]

Question 11 (Case Study Analysis) [4 Marks]

Mira designs a target game using a circular cardboard of radius $R = 14$ cm with center O. The cardboard is divided into three distinct concentric regions:
· Region X (Inner Bullseye): A central circle of radius $r_1 = 7$ cm.
· Region Y (Middle Ring): An annular ring between radius $r_1 = 7$ cm and $r_2 = 14$ cm.
· Region Z (Semicircular Sector): A half-annulus formed by a diameter AB cutting across Region Y.

A pellet is dropped at random onto the circular board. Taking $\pi = \frac{22}{7}$, answer the following:



Geometric Formulas: Total cardboard disc has radius $R = 14$ cm. Inner circle has radius $r_1 = 7$ cm.

(i) Total area of the complete circular cardboard disc [1 Mark]

$A_{\text{total}} = \pi R^2 = \frac{22}{7} \times 14^2 = \frac{22}{7} \times 196 = 22 \times 28 = 616 \text{ cm}^2$.

Answer (i): Total Area = 616 cm²

(ii) Area of Region X (inner bullseye circle) [1 Mark]

$A_X = \pi r_1^2 = \frac{22}{7} \times 7^2 = 22 \times 7 = 154 \text{ cm}^2$.

Answer (ii): Area of Region X = 154 cm²

(iii) Probability that the pellet lands in Region X [1 Mark]

$P(\text{Region X}) = \frac{\text{Area of Region X}}{\text{Total Area}} = \frac{154}{616} = \frac{1}{4} = 0.25$.

Answer (iii): P(Region X) = $\frac{1}{4}$ = 0.25

(iv) Probability that the pellet lands in Region Y (the annular ring) [1 Mark]

Area of Region Y $= A_{\text{total}} - A_X = 616 - 154 = 462 \text{ cm}^2 \implies P(\text{Region Y}) = \frac{462}{616} = \frac{3}{4} = 0.75$.

Answer (iv): P(Region Y) = $\frac{3}{4}$ = 0.75

— END OF WORKED SOLUTIONS (TEST–10) · MATHS MANTHAN TUITION CLASSES —